CATALYST. BY MRS. RANA · GRADE 9 · SCIENCE · TERM 1

Term Glossary

Every term of the term in English and Arabic. Tick the box when you know it.

Term	بالعربي	Meaning
MODULE 2 · LESSON 1 · ORGANISMS AND THEIR RELATIONSHIPS الكائنات الحية وعلاقاتها		
☐ **Ecology**	علم البيئة	The study of the relationships among living organisms and the interactions they have with their environment.
☐ **Ecologist**	عالم بيئة	A scientist who observes, experiments and models to study organisms and their environments.
☐ **Biosphere**	المحيط الحيوي	The portion of Earth that supports life, a thin layer from the atmosphere to the deep ocean.
☐ **Scientific model**	نموذج علمي	A visual, mathematical or computer representation used to answer a question or test a hypothesis.
☐ **Computer model**	نموذج حاسوبي	A simulation run with software to test different ways of solving a problem.
☐ **Mathematical model**	نموذج رياضي	An equation used to represent and predict a natural process.
☐ **Biotic factor**	عامل حيوي	A living factor in an organism's environment, such as fish, algae or frogs.
☐ **Abiotic factor**	عامل غير حيوي	A nonliving factor in an organism's environment, such as temperature, sunlight or soil.
☐ **Limiting factor**	عامل محدد	Any factor that restricts the numbers, reproduction or distribution of organisms.
☐ **Tolerance**	التحمّل	The ability of an organism to survive when subjected to abiotic or biotic factors.
☐ **Optimum range**	المدى الأمثل	The part of the range where conditions are best and the most organisms live.
☐ **Zone of physiological stress**	منطقة الإجهاد الفسيولوجي	Conditions where organisms survive but may be unable to grow or reproduce.
☐ **Zone of intolerance**	منطقة عدم التحمّل	Conditions beyond the upper or lower limits where the organisms die.
☐ **Population**	الجماعة	Individual organisms of a single species that share the same location at the same time.
☐ **Biological community**	المجتمع الحيوي	A group of interacting populations that occupy the same area at the same time.
☐ **Ecosystem**	النظام البيئي	A community of living things together with all the nonliving factors that affect it.
☐ **Biome**	البيوم	A large group of ecosystems that share the same climate and have similar types of communities.
☐ **Organism**	الكائن الحي	An individual living thing; the lowest level of organization.
☐ **Levels of organization**	مستويات التنظيم	Smaller pieces of the biosphere studied by ecologists, increasing in complexity.
☐ **Habitat**	الموطن	The area where an organism lives.
☐ **Niche**	العش البيئي	The role or position an organism has in its environment: how it meets its needs for food, shelter and reproduction.
☐ **Competition**	التنافس	When more than one organism uses a resource at the same time.
☐ **Predation**	الافتراس	The act of one organism pursuing and consuming another organism for food.
☐ **Symbiosis**	التكافل	The close relationship that exists when two or more species live together.
☐ **Mutualism**	تبادل المنفعة	A relationship between organisms that live closely together and both gain.
☐ **Commensalism**	التعايش	A relationship in which one organism gains and the other is neither helped nor harmed.

Term	بالعربي	Meaning
☐ **Parasitism**	التطفل	A relationship in which one organism gains at the expense of another, its host.
MODULE 2 · LESSON 2 · FLOW OF ENERGY IN AN ECOSYSTEM تدفق الطاقة في النظام البيئي		
☐ **Autotroph**	ذاتي التغذية	An organism that collects energy from sunlight or inorganic substances to produce food.
☐ **Heterotroph**	غير ذاتي التغذية	An organism that gets its energy requirements by consuming other organisms; a consumer.
☐ **Herbivore**	آكل النباتات	A consumer that eats only plants, such as a cow or rabbit.
☐ **Carnivore**	آكل اللحوم	A consumer that preys on other consumers, such as a wolf or lion.
☐ **Omnivore**	القارت	A consumer that eats both plants and animals, such as a bear or human.
☐ **Detritivore**	آكل الفتات	An organism that eats fragments of dead matter, returning nutrients to soil, air and water.
☐ **Decomposer**	المحلل	An organism such as a fungus that breaks down dead organisms by releasing digestive enzymes.
☐ **Food chain**	السلسلة الغذائية	A simple model that shows a single path of energy flow from organism to organism.
☐ **Food web**	الشبكة الغذائية	A model representing many interconnected pathways in which energy flows through a group of organisms.
☐ **Trophic level**	المستوى الغذائي	Each step in a food chain or food web.
☐ **Ecological pyramid**	الهرم البيئي	A diagram showing relative amounts of energy, biomass or numbers of organisms at each step of feeding.
☐ **Biomass**	الكتلة الحية	The total mass of living matter at each trophic level.
☐ **Pyramid of numbers**	هرم الأعداد	A diagram showing how many organisms are found at each feeding step.
MODULE 2 · LESSON 3 · CYCLING OF MATTER دورة المادة		
☐ **Nutrient**	عنصر غذائي	A chemical substance that an organism must obtain from its environment to sustain life.
☐ **Biogeochemical cycle**	الدورة البيوجيوكيميائية	The exchange of matter through the biosphere involving living, geological and chemical processes.
☐ **Matter**	المادة	Anything that takes up space and has mass.
☐ **Evaporation**	التبخر	Liquid water turning into water vapor using energy from the Sun.
☐ **Transpiration**	النتح	Water vapor leaving the surfaces of plants (about 10% of water vapor).
☐ **Condensation**	التكاثف	Water vapor cooling into droplets around dust particles to form clouds.
☐ **Precipitation**	الهطول	Water falling from clouds to Earth's surface as rain or snow.
☐ **Percolation**	التسرب	Surface water moving down through the soil into groundwater.
☐ **Photosynthesis**	البناء الضوئي	The process in which plants and algae convert carbon dioxide and water into carbohydrates and release oxygen.
☐ **Cellular respiration**	التنفس الخلوي	The process in which living organisms consume oxygen and release carbon dioxide.
☐ **Fossil fuels**	الوقود الأحفوري	Coal, oil or gas formed when organic matter is buried underground for a long time.

Term	بالعربي	Meaning
☐ **Combustion**	الاحتراق	Burning fossil fuels, which returns carbon dioxide to the atmosphere.
☐ **Calcium carbonate**	كربونات الكالسيوم	A compound of Ca, C and O that forms shells and limestone rock (chalk).
☐ **Decay**	التحلل	The breakdown of dead organisms, which releases CO_2 to the atmosphere.
☐ **Nitrogen fixation**	تثبيت النيتروجين	The conversion of N_2 gas into ammonium (NH_4^+) by specialized bacteria.
☐ **Nitrification**	النترجة	The chemical process that turns ammonium into NO_2^- and NO_3^-.
☐ **Assimilation**	التمثيل (الامتصاص)	Plants absorbing nitrate through their roots and using it to build proteins.
☐ **Denitrification**	عكس النترجة	The process in which soil bacteria convert NO_2^- and NO_3^- back into N_2 gas.
☐ **Algae bloom**	تكاثر الطحالب	Overgrowth of tiny water producers caused by too much nitrogen or phosphorus in water.
☐ **Root nodules**	العقد الجذرية	Swellings on the underground parts of legumes where bacteria that fix nitrogen live.
☐ **Acid rain**	المطر الحمضي	Precipitation formed when nitrogen-oxygen compounds from factories combine with water in the air.
☐ **Phosphorus**	الفوسفور	An element essential for the growth and development of organisms, cycled through rock, soil, water and organisms.
☐ **Sedimentation**	الترسيب	Settling of phosphate in water to form sediment and, over time, rock.
☐ **Uplift**	الرفع الجيولوجي	Geological change that moves sediment from water back up onto land.
☐ **Weathering**	التجوية	The slow breakdown of rocks that adds phosphate back to the soil.
☐ **Phosphate mining**	تعدين الفوسفات	Digging P-rich rock out of the ground, a human activity that increases the P cycling through the environment.
MODULE 5 · LESSON 1 · BIODIVERSITY التنوع الحيوي		
☐ **Biodiversity**	التنوع الحيوي	The variety of life in an area that is determined by the number of different species in that area.
☐ **Extinction**	الانقراض	The permanent disappearance of a species when its last member dies.
☐ **Genetic diversity**	التنوع الجيني	The variety of genes or inheritable characteristics present in a population.
☐ **Species diversity**	تنوع الأنواع	The number of different kinds and the relative abundance of each kind in a biological community.
☐ **Ecosystem diversity**	تنوع الأنظمة البيئية	The variety of habitats and communities present in the biosphere.
☐ **Direct economic value**	القيمة الاقتصادية المباشرة	Benefit from products humans use: food, clothing, energy, medicine and shelter.
☐ **Indirect economic value**	القيمة الاقتصادية غير المباشرة	Benefit from services of a healthy biosphere, such as oxygen, clean water and fertile soil.
☐ **Watershed**	مستجمع المياه	A land area where the water on it or under it drains to the same place.
☐ **Aesthetic value**	القيمة الجمالية	The worth of landscapes that give recreation or inspiration.
☐ **Genetic engineering**	الهندسة الوراثية	Transferring the units that control inherited traits from one species to another.
☐ **Teosinte**	التيوسنت	A wild relative of corn with genes that resist viral diseases.

Term	بالعربي	Meaning
MODULE 5 · LESSON 2 · THREATS TO BIODIVERSITY التهديدات للتنوع الحيوي		
☐ **Background extinction**	الانقراض الخلفي	The gradual, natural process of species dying out over time.
☐ **Mass extinction**	الانقراض الجماعي	An event in which a large percentage of all living species die out in a relatively short period of time.
☐ **Extinction**	الانقراض	When the last member of a species dies and it is lost from the biosphere forever.
☐ **Paleontologist**	عالم الحفريات	A scientist who studies fossils of organisms that lived long ago.
☐ **Island species**	أنواع الجزر	Organisms on isolated land in water, especially vulnerable: no natural predators, small populations.
☐ **Anthropogenic change**	تغيير بشري المنشأ	A change to the environment induced by human activity.
☐ **Natural resources**	الموارد الطبيعية	All materials and organisms found in the biosphere, such as minerals, fuels, soil, clean water and solar energy.
☐ **Overexploitation**	الاستغلال المفرط	The excessive use of species that have economic value.
☐ **Keystone species**	النوع المفتاحي	A species that plays such a large role in an ecosystem that its decline affects many others.
☐ **Habitat fragmentation**	تجزئة الموطن	The separation of an ecosystem into small pieces of land.
☐ **Edge effects**	تأثيرات الحافة	Different environmental conditions that occur along the boundaries of an ecosystem.
☐ **Biological magnification**	التضخم الحيوي	The increasing concentration of toxic substances in organisms as trophic levels increase.
☐ **Eutrophication**	الإثراء الغذائي	Water pollution when nitrogen- and phosphorus-rich substances cause extensive algae growth that uses up oxygen.
☐ **Acid precipitation**	الهطول الحمضي	Rain, sleet, snow or fog carrying sulfuric or nitric compounds that come from burned fuels.
☐ **Invasive species**	الأنواع الغازية	Introduced organisms that significantly modify or disrupt their new habitat.
☐ **Introduced species**	الأنواع المُدخلة	Organisms transported to a new habitat, on purpose or by accident.
☐ **DDT**	مبيد DDT	A pesticide (1940s–1970s) that builds up in tissues and caused thin eggshells in fish-eating birds.
MODULE 5 · LESSON 3 · CONSERVING BIODIVERSITY الحفاظ على التنوع الحيوي		
☐ **Renewable resource**	مورد متجدد	A resource replaced by natural processes faster than it is consumed, such as solar energy.
☐ **Nonrenewable resource**	مورد غير متجدد	A resource found in limited amounts or replaced over extremely long periods, such as coal.
☐ **Sustainable use**	الاستخدام المستدام	Using resources at a rate at which they can be replaced or recycled while preserving the biosphere's long-term health.
☐ **Conservation**	الحفاظ	Reducing the amount consumed, recycling and preserving ecosystems while using them responsibly.
☐ **Standard of living**	مستوى المعيشة	The available amount of goods and money in a country.
☐ **Endemic species**	الأنواع المتوطنة	Organisms that are only found in one specific geographic area.
☐ **Biodiversity hotspot**	النقطة الساخنة للتنوع	A region with at least 1500 endemic vascular plant species that has lost at least 70% of its original habitat.
☐ **Corridor**	الممر البيئي	A passageway between habitat fragments that lets organisms move safely.

Term	بالعربي	Meaning
☐ **Buffer zone**	المنطقة العازلة	An area around a protected zone where sustainable use of natural resources is permitted.
☐ **Megareserve**	المحمية الكبرى	A large managed area with protected zones surrounded by zones of sustainable use (Costa Rica).
☐ **CITES**	اتفاقية سايتس	A 1973 international treaty that outlawed trade in endangered species and animal parts.
☐ **Bioremediation**	المعالجة الحيوية	The use of living organisms such as prokaryotes, fungi or plants to detoxify a polluted area.
☐ **Biological control**	المكافحة الحيوية	Introducing a natural predator or competitor to manage an invasive species.
☐ **Hydrilla**	الهيدريلا	An invasive water plant in Florida that forms dense mats blocking light.
☐ **Recovery time**	زمن التعافي	How long a community takes to come back after a disaster; depends on area size and disturbance type.
MODULE 1 · LESSON 1 · THE METHODS OF SCIENCE طرق العلم		
☐ **Science**	العلم	A process based on inquiry that helps develop explanations about events in nature (Latin: knowledge).
☐ **Life science**	علوم الحياة	The branch that deals with living things.
☐ **Earth science**	علوم الأرض	The branch that investigates our planet and space.
☐ **Physical science**	العلوم الفيزيائية	The branch that studies matter and energy.
☐ **Electron cloud**	السحابة الإلكترونية	The space around the nucleus where tiny negative particles move rapidly.
☐ **Investigation**	التحقيق العلمي	A way scientists learn: observing, experimenting with a control, or modeling.
☐ **Scientific methods**	الطرق العلمية	The pattern of investigation procedures scientists often follow.
☐ **Hypothesis**	الفرضية	A possible answer to a question or solution to a problem, based on what you know and observe.
☐ **Experiment**	التجربة	A test of the effect of one thing on another using a control.
☐ **Variable**	المتغير	A quantity that can have more than a single value.
☐ **Independent variable**	المتغير المستقل	The factor changed by the scientist to test its effect.
☐ **Dependent variable**	المتغير التابع	The factor that changes in response to the changed factor; the one measured.
☐ **Constant**	الثابت	A factor that does not change when other variables change.
☐ **Control**	الضابط (المجموعة الضابطة)	The standard by which test results can be compared.
☐ **Peer review**	مراجعة الأقران	A process in which procedures and results are evaluated by other scientists in the same field.
☐ **Bias**	التحيز	When a scientist's expectations change how results are analyzed or conclusions are formed.
☐ **Blind experiment**	التجربة العمياء	A test in which researchers don't know which group is the study group and which is the control.
☐ **Model**	النموذج	A representation of an idea, event or object to help people better understand it.
☐ **Computer simulation**	المحاكاة الحاسوبية	Software that safely mimics a process to test it and collect data.

Term	بالعربي	Meaning
☐ **Scientific theory**	النظرية العلمية	An explanation based on knowledge gained from many observations and investigations.
☐ **Scientific law**	القانون العلمي	A statement about what happens in nature that seems to be true all the time.